

Beer–Lambert Absorption Enhancement in Structural and Material Slow-Light

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Session T2B, Electromagnetic Wave Propagation



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**Photonics and
Electromagnetics Group**

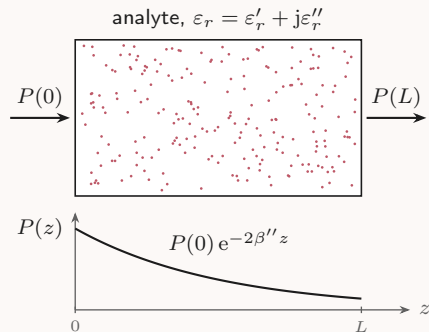
- ▶ Transmitted power through an absorbing medium:

$$P(z) = P(0) e^{-\alpha z}, \quad \alpha = 2\beta'',$$

for propagation constant $\beta = \beta' + j\beta''$.

- ▶ Bouguer (1729), Lambert (1760): attenuation is exponential in path length. Beer (1852): α is linear in concentration [A. Beer, *Ann. Phys.* **162**, 78–88 (1852)].
- ▶ **Analyte**: absorbing gas, liquid, or dilute component,

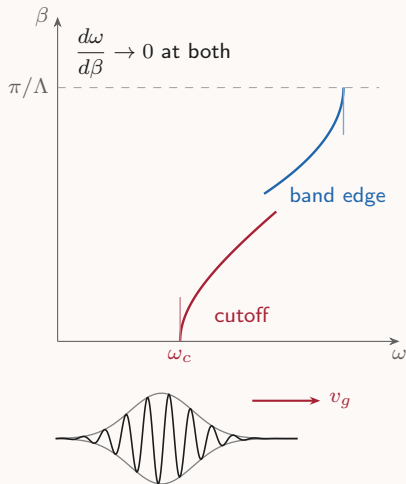
$$\varepsilon_r = \varepsilon'_r + j\varepsilon''_r, \quad \varepsilon''_r > 0 \text{ represents loss.}$$



- ▶ For a dispersion relation $F(\omega, \beta) = 0$, define group velocity and index:

$$n_g = c \operatorname{Re} \left[\frac{d\beta}{d\omega} \right], \quad v_g = \frac{c}{n_g}.$$

- ▶ **Slow light:** $n_g \gg 1$ (flat $\omega(\beta)$), from
 - ▶ **material** dispersion: atomic lines, EIT, Raman, Brillouin;
 - ▶ **structural** (waveguide design): band edges, resonators, cutoff.
- ▶ **Operational meaning** [J. Peatross et al., Phys. Rev. Lett. 84, 2370–2373 (2000)]: with real ω and complex $\beta(\omega)$, the Poynting-flux centroid delay over Δz is exactly $\langle \operatorname{Re}[d\beta/d\omega] \rangle \Delta z$, averaged over the output spectrum, plus a reshaping term that vanishes with the bandwidth.



Does slow light necessarily enhance absorption?

- ▶ [S. Chin et al., Adv. Opt. Sci. Congress (2009)]
[L. Thévenaz et al., Adv. Slow Fast Light V (2012)]
[I. Dicaire et al., Opt. Lett. **37**, 4934–4936 (2012)]: **measured**: no enhancement in a Brillouin-slowed solid-core fibre with acetylene in its holes; enhancement in a structural photonic-crystal guide.
- ▶ [J. Grgić et al., Opt. Express **18**, 14270–14279 (2010)]: **computed**: photonic-crystal-fibre enhancement does not exactly follow n_g .
- ▶ [H. G. Winful, Phys. Rep. **436**, 1–69 (2006)]: for fixed-length evanescent systems, group delay is a dwell time, not a transit time.
- ▶ [E. O. Schulz-DuBois, Proc. IEEE **57**, 1748–1757 (1969)]: **energetic interpretation** propagating wave: $2\beta'' = \gamma_D/v_E$; v_E : energy velocity, γ_D : loss per stored energy.

Is n_g the index that measures absorption enhancement?

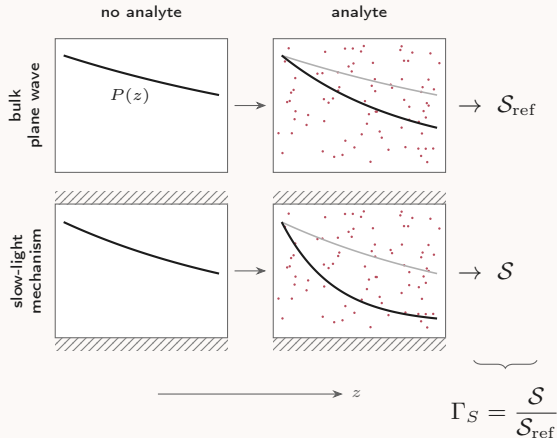
Loss sensitivity and its enhancement

- **Loss sensitivity** at fixed ω , geometry, ε'_r , ($k_0 = \omega/c$):

$$\mathcal{S} = \frac{1}{k_0} \frac{\partial \beta''}{\partial \varepsilon_r''}.$$

- **Reference** $\mathcal{S}_{\text{ref}}$: plane wave without slow light (structural: **analyte alone**; material: **gain line off**).
- **Enhancement factor**: $\Gamma_S = \mathcal{S}/\mathcal{S}_{\text{ref}}$.

Remark : One metric for **any linear slow-light mechanism** (any enhancement mechanism, more in general) with complex β .



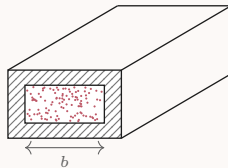
One dispersion relation, two physical cases

- ▶ $n = n' + jn''$: uniform, non-dispersive filling index; $\varepsilon_r = n^2$.
- ▶ $k_{\perp}$: TE/TM transverse wavenumber; $n_{\perp} = k_{\perp}/k_0$.
For PEC TE₀₁, $k_{\perp} = \pi/b$.

For $\mathbf{E} = \psi(x, y) e^{j(\beta z - \omega t)}$ and $\nabla_{\perp}^2 \psi = -k_{\perp}^2 \psi$,

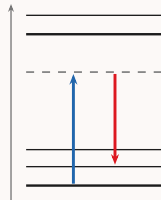
$$\beta^2 = k_0^2 n^2 - k_{\perp}^2.$$

$k_{\perp} \neq 0$: structural



PEC walls fix $k_{\perp}$;
analyte-filled guide.

$k_{\perp} = 0$: material



Plane wave; dispersion
entirely in $n(\omega)$.

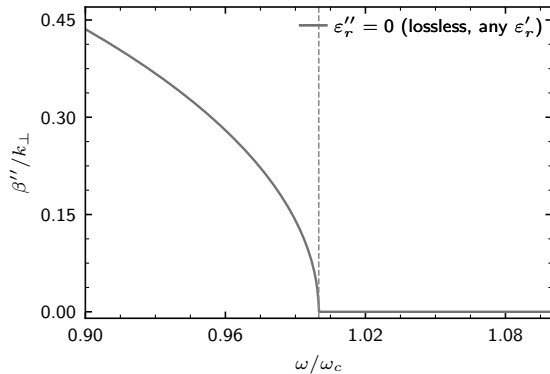
Analyte-filled PEC guide, TE mode:

$$n_{\text{eff}} = \frac{\beta}{k_0} = \sqrt{\epsilon_r - n_{\perp}^2}.$$

Cutoff. Lossless filling: $\beta' = \beta'' = 0$ at $k_0^2 \epsilon'_r = k_{\perp}^2$,

$$\omega_c = \frac{c k_{\perp}}{\sqrt{\epsilon'_r}} \quad \left(\text{TE}_{01} : \omega_c = \frac{\pi c}{b \sqrt{\epsilon'_r}} \right).$$

$\omega > \omega_c$: propagating; $\omega < \omega_c$: **evanescent**, $\beta'' > 0$ **even without loss**. With loss, β remains complex through cutoff.



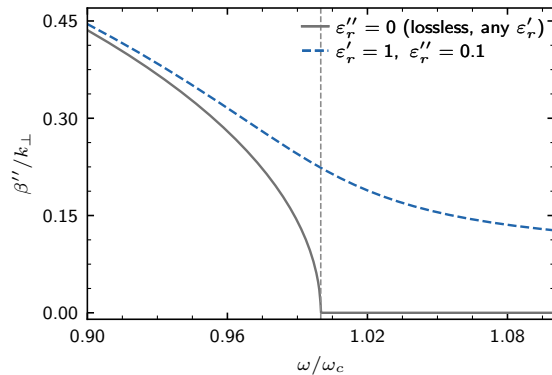
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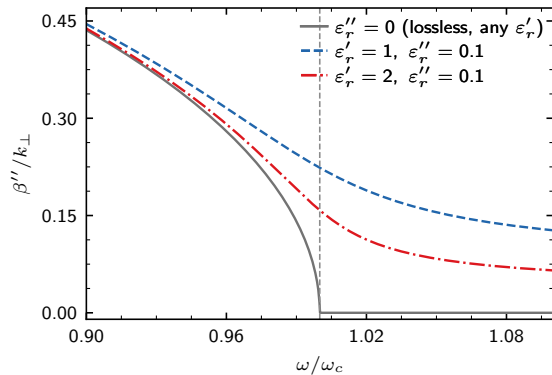
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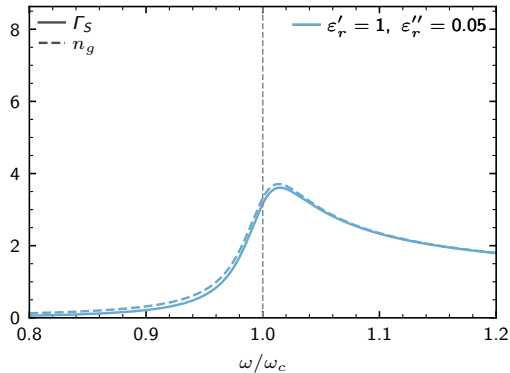


Non-dispersive filling ($dn_{\perp}/d\omega = -n_{\perp}/\omega$):

$$n_g^{\text{wg}} = n'_{\text{eff}} + \omega \frac{dn'_{\text{eff}}}{d\omega} = n'_{\text{eff}} \left(1 + \frac{n_{\perp}^2}{|n_{\text{eff}}|^2} \right),$$

$\mathcal{S}^{\text{wg}} = n'_{\text{eff}}/(2|n_{\text{eff}}|^2)$, $\mathcal{S}_{\text{ref}} = n'_{\text{ref}}/(2|n_{\text{ref}}|^2)$ (plane wave, same ε_r).

	$\omega \ll \omega_c$	$\omega \gg \omega_c$
	evanescent limit	plane-wave limit
n_{eff}	$\rightarrow jn_{\perp}$	$\rightarrow n$
$\beta''/k_{\perp}$	$\rightarrow 1$	$\rightarrow k_0 n''/k_{\perp}$
n_g	$\rightarrow 0$	$\rightarrow n'$
Γ_S	$\rightarrow 0$	$\rightarrow 1$

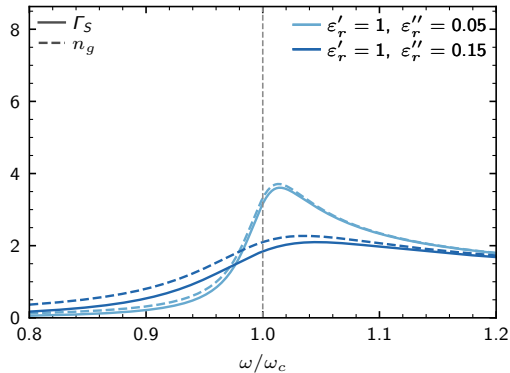


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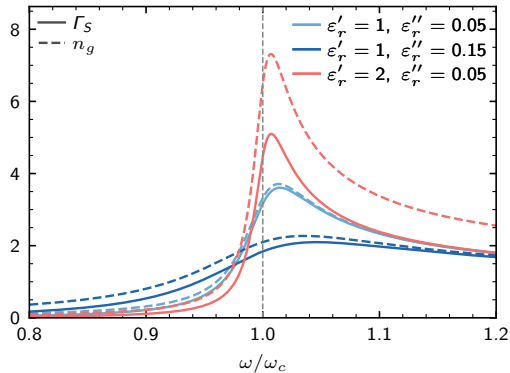


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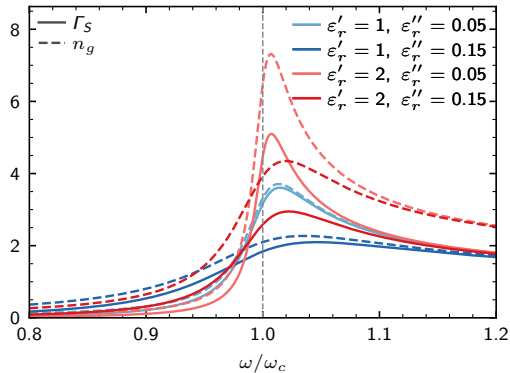


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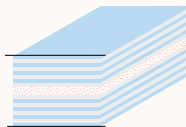
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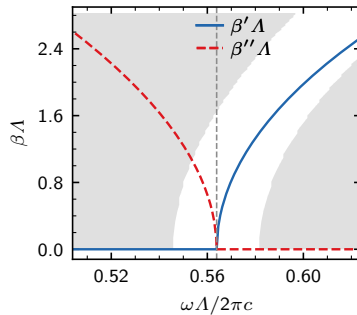


A photonic-crystal waveguide near cutoff



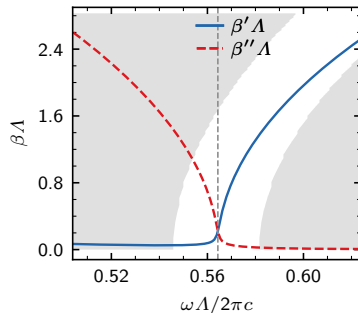
Analyte core between Bragg claddings (N periods $A|B$): the stop band replaces the PEC wall. The core mode has a **cutoff** ω_c ($\beta' = \beta'' = 0$) in the gap.

ideal, semi-infinite cladding



Square-root cutoff at ω_c .

finite, eight periods

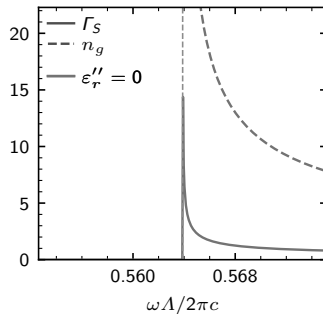


Radiation loss modifies the cutoff.

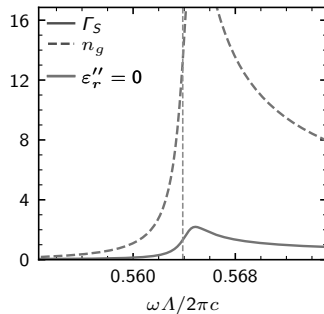
Example: $\Lambda = 500$ nm; $(d_A, d_B) = (250, 250)$ nm; $(n_A, n_B) = (1.45, 2.10)$; air core 669 nm; TE even mode. Second-order gap: $\omega_c \Lambda / 2\pi c = 0.564$, or $\omega_c = 2\pi \times 338.1$ THz.

- ▶ Ideal, lossless: Γ_S diverges at cutoff; core loss limits the peak.
- ▶ Relative to PEC: reduced by the energy fraction in the analyte.
- ▶ Finite cladding: radiation also limits the peak.

ideal, semi-infinite cladding



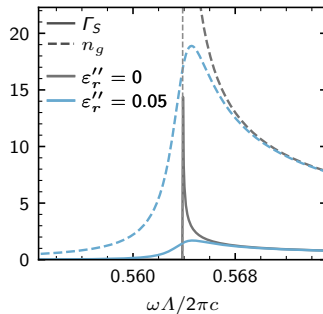
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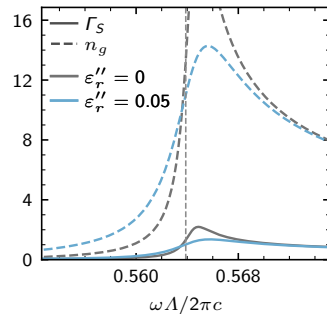
Loss sensitivity rises near cutoff, but is not equal to n_g

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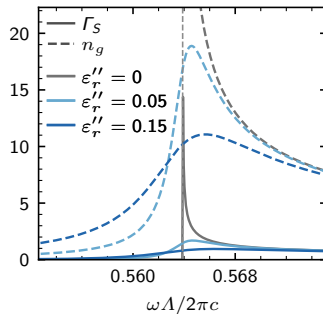


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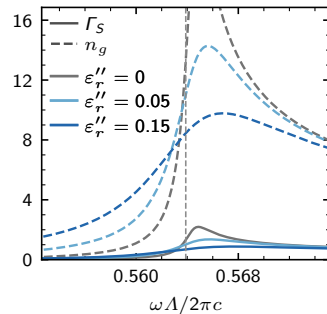


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ideal, semi-infinite cladding



finite, eight periods



Filling fraction: lossless electric-energy fraction in the analyte core,

$$\eta_{\text{fill}} = \frac{\varepsilon_c \int_{\text{core}} |E|^2 dx}{\int \varepsilon(x) |E|^2 dx},$$

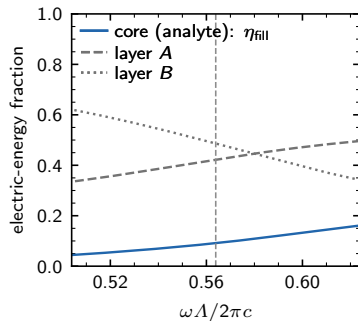
ε_c : core; $\varepsilon(x)$: transverse profile; x : transverse coordinate.

Normalised enhancement
(plane-wave loss $\eta_{\text{fill}} \varepsilon_r''$):

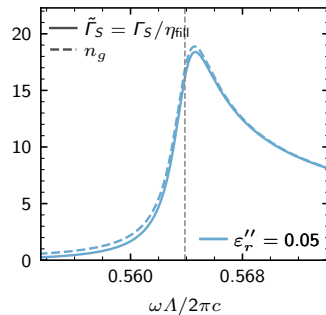
$$\tilde{\Gamma}_S = \frac{\Gamma_S}{\eta_{\text{fill}}} = \frac{\mathcal{S}}{\eta_{\text{fill}} \mathcal{S}_{\text{ref}}}.$$

► Lossless closed guide:

$$\tilde{\Gamma}_S = n_g / n_{\text{core}}.$$



ideal cladding



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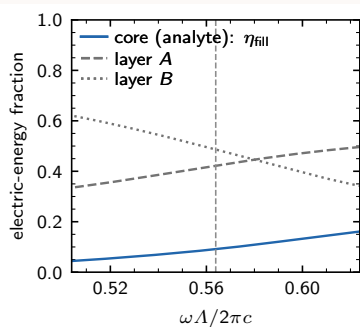
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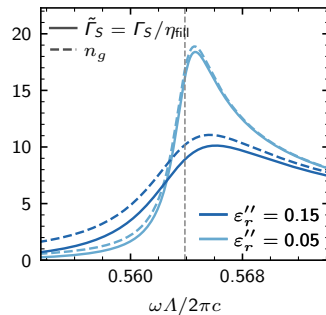
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ideal cladding



Plane wave ($k_{\perp} = 0$): background

$\varepsilon_B = \varepsilon_0(\varepsilon'_{B,r} + j\varepsilon''_{B,r})$, Lorentzian gain, and analyte:

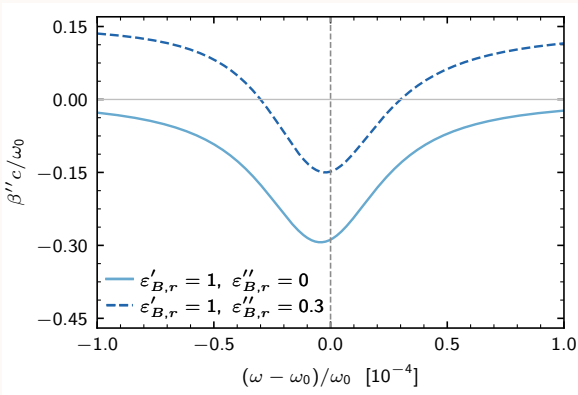
$$\varepsilon(\omega) = \varepsilon_B + \varepsilon_0 \frac{G}{\omega - \omega_0 + j\gamma_R} + j\varepsilon_0 \varepsilon''_{A,r}.$$

- **background**: non-dispersive, with loss $\varepsilon''_{B,r}$;
- **gain** (G, γ_R): pump-controlled slow light;
- **analyte** $\varepsilon''_{A,r}$: differentiation variable only.

$$n_g^{\text{pw}} = n' + \omega \frac{dn'}{d\omega}, \quad \frac{dn}{d\omega} = -\frac{(n^2 - n_B^2)^2}{2Gn},$$

$$n_B^2 = \varepsilon_B / \varepsilon_0, \quad \mathcal{S}^{\text{pw}} = n' / (2|n|^2).$$

Example: $\gamma_R = 3 \times 10^{-5} \omega_0$ (5.8 GHz half-width at $1.55 \mu\text{m}$), $G/\gamma_R = 0.6$, set deliberately high [Y. Okawachi et al., Opt. Express 14, 2317–2322 (2006)].



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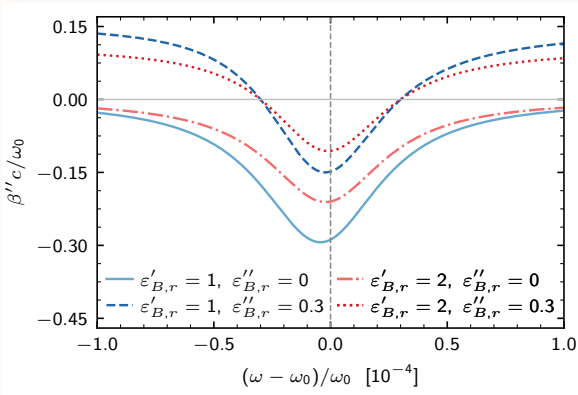
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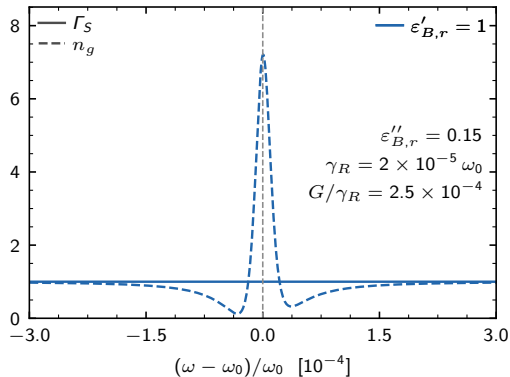
- ▶ Narrow peak of n_g at the line centre.
- ▶ Both backgrounds: $|\Gamma_S - 1| < 10^{-4}$ over the plotted interval.

Experimental evidence

Brillouin slow light in a solid-core photonic-crystal fibre, acetylene in the holes: **no enhancement observed.**

[S. Chin et al., Adv. Opt. Sci. Congress (2009)]

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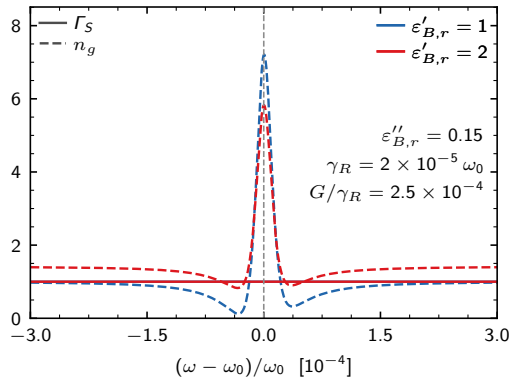
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Why n_g is not a universal enhancement metric

Configuration	Origin of dispersion	What sets n_g	What sets $\mathcal{S}$	Result
PEC guide	waveguide cutoff	$1/ n_{\text{eff}} $	$1/ n_{\text{eff}} $	$\Gamma_S \simeq n_g/n'$ (low loss)
Photonic-crystal guide	waveguide cutoff (modified by leakage)	$1/ n_{\text{eff}} $	$\eta_{\text{fill}}/ n_{\text{eff}} $	$\Gamma_S \simeq \eta_{\text{fill}}n_g/n_{\text{core}}$ (low loss/leakage)
Lorentz gain line	$\varepsilon(\omega)$ resonance	$dn'/d\omega$	$ n $	$\Gamma_S \simeq 1$ (line model)

1. **Energy balance with evanescence.** Extend $2\beta'' = \gamma_D/v_E$ to reactive stored energy through cutoff.
[E. O. Schulz-DuBois, Proc. IEEE **57**, 1748–1757 (1969)]
[G. Diener, Phys. Lett. A **235**, 118–124 (1997)]
2. **Open waveguides.** Separate absorption from radiation (for example in the ARROW case). Can $\partial\beta''/\partial\epsilon_r''$ be negative?
3. **Other material resonances.** Test $|\Gamma_S - 1| \ll 1$ for EIT, Fano, and gain–loss media at $\mathcal{PT}$ -symmetry breaking.
4. **Nonlinear enhancement.** Define a separate metric for processes governed by energy density and phase matching. [V. Eckhouse et al., Opt. Lett. **35**, 1440–1442 (2010)]

Conclusions

1. We introduce $\mathcal{S} = \partial\beta''/(k_0\partial\varepsilon_r'')$ and $\Gamma_S = \mathcal{S}/\mathcal{S}_{\text{ref}}$ to quantify Beer–Lambert enhancement.
2. Near a structural cutoff the mode is not rigorously but partly evanescent, and the same analyte loss coefficient produces more attenuation.
3. At low core loss and leakage, $\Gamma_S \simeq \eta_{\text{fill}}n_g/n_{\text{core}}$.



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Thank you for your attention

Questions?

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Part A

Additional material

Write $\beta^2 = k_0^2 n^2 - k_\perp^2$ in real and imaginary parts:

$$\beta'^2 - \beta''^2 = k_0^2 (n'^2 - n''^2 - n_\perp^2), \quad \beta' \beta'' = k_0^2 n' n''.$$

Eliminate β' :

$$\beta''^2 = \frac{1}{2} k_0^2 \left[\sqrt{(n'^2 - n''^2 - n_\perp^2)^2 + 4n' n''} - (n'^2 - n''^2 - n_\perp^2) \right].$$

Normalisations

Non-magnetic, $\varepsilon = \varepsilon_0 n^2$: $\varepsilon'_r = n'^2 - n''^2$, $\varepsilon''_r = 2n' n''$. Frequencies are divided by ω_c (waveguide), ω_0 (material line) or c/Λ (photonic crystal); β'' by $k_\perp$ or ω_0/c .

Transfer-matrix dispersion relation of the N -period guide

Planar TE, half of the structure by symmetry: core of width w , N periods AB , then the exterior. In layer j (index n_j , thickness d_j):

$$q_j = \sqrt{n_j^2 k_0^2 - \beta^2}, \quad M_j = \begin{pmatrix} \cos q_j d_j & \frac{\sin q_j d_j}{q_j} \\ -q_j \sin q_j d_j & \cos q_j d_j \end{pmatrix}.$$

One period: $M = M_B M_A$. N periods: $M^N = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$.

Core edge

$$z_c = \begin{cases} -q_c \tan \frac{q_c w}{2} & \text{even} \\ q_c \cot \frac{q_c w}{2} & \text{odd} \end{cases}$$

Exterior

$$z_e = j q_e, \quad \operatorname{Re} q_e \geq 0 \\ (\text{radiation condition})$$

Cladding

$$z_N = \frac{z_e A - C}{D - z_e B}$$

$$z_c(\omega, \beta) - z_N(\omega, \beta) = 0$$

The same equation for every model; only z_N changes: hard wall (PEC), N periods, or the Bloch limit $N \rightarrow \infty$, in which the cladding is the bulk crystal, $\cos K\Lambda = \Delta = \frac{1}{2} \operatorname{tr} M$. Bound modes exist only in its stop bands, $|\Delta| > 1$; a finite N turns them into leaky modes.

- [1] A. Beer, Ann. Phys. **162**, 78 (1852).
- [2] J. Peatross, S. A. Glasgow, and M. Ware, Phys. Rev. Lett. **84**, 2370 (2000).
- [3] S. Chin et al., in Adv. Opt. Sci. Congress (2009), SMA3.
- [4] L. Thévenaz, I. Dicaire, and S. Chin, in Adv. Slow Fast Light V, Vol. 8273 (2012), p. 82731D.
- [5] I. Dicaire et al., Opt. Lett. **37**, 4934 (2012).
- [6] J. Grgić et al., Opt. Express **18**, 14270 (2010).
- [7] H. G. Winful, Phys. Rep. **436**, 1 (2006).
- [8] E. O. Schulz-DuBois, Proc. IEEE **57**, 1748 (1969).
- [9] Y. Okawachi et al., Opt. Express **14**, 2317 (2006).
- [10] G. Diener, Phys. Lett. A **235**, 118 (1997).
- [11] V. Eckhouse et al., Opt. Lett. **35**, 1440 (2010).